

Equations for NE533

$$\frac{\delta^n}{2} = K^n t \rightarrow \delta = K t^{1/n}$$

$n \approx 4$ for Zr alloys

$$\delta^*(\mu m) = 5.1 \exp\left(\frac{-550}{T}\right)$$

$$t^*(d) = 6.62 \times 10^{-7} \exp\left(\frac{11949}{T}\right)$$

$$K_L\left(\frac{\mu m}{d}\right) = 7.48 \times 10^6 \exp\left(\frac{-12500}{T}\right)$$

$$\delta(\mu m) = \delta^* + K_L(t - t^*)$$

$$w = (m_2 - m_1)/S \quad (mg/dm^2)$$

$$\delta(\mu m) = \frac{w}{14.7}$$

$$C_H^{clad}(wt.ppm) = \frac{2f \times \delta \times \rho_{oxide} \times f_{ZrO_2}^O \times \frac{M_H}{M_O}}{(t - \frac{\delta}{P_{BR}}) \times \rho_{metal}} \times 10^6$$